

CENG381 Stochastic Processes

Gambler's Ruin — Practice 1

Q1 (30 points). Alice and Bob play a series of fair coin-flip rounds. Alice starts with \$3 and Bob starts with \$5 (total stake \$8). Each round, Alice wins \$1 (prob 1/2) or loses \$1 (prob 1/2). The game ends when one player is bankrupt.

- Set up the Gambler's Ruin model on states $\{0, 1, \dots, 8\}$ with $p_0 = 0$ (Alice bankrupt) and $p_8 = 1$ (Bob bankrupt). Write the recurrence relation for $p_k = P(\text{Alice wins} \mid \text{starts with } \$k)$.
 - Derive $p_k = k/n$ by solving the recurrence. Show that the solution $p_k = k \cdot p_1$ combined with $p_n = 1$ gives $p_1 = 1/n$ and hence $p_k = k/n$. Compute p_3 .
 - What is the probability that Alice loses (goes bankrupt)? If Alice's stake doubles to \$6 (with Bob at \$2), how does her win probability change?
-

Q2 (35 points). In the Gambler's Ruin game on $\{0, 1, \dots, n\}$ with absorbing barriers at 0 and n , let f_k denote the expected number of rounds until the game ends, starting from position k .

- Write the recurrence relation for f_k (for $1 \leq k \leq n-1$) and state the boundary conditions.
 - Verify that $f_k = k(n-k)$ satisfies the recurrence and boundary conditions. Carry out the algebra explicitly.
 - For $n = 8$ and $k = 3$, compute $E[\text{game length}]$. Which starting position k^* maximises the expected game length? Compute this maximum.
-

Q3 (35 points). A drunk walker starts at position $k = 2$ on the integer path $\{0, 1, 2, 3, 4, 5\}$ ($n = 5$). State 0 = 'falls in a ditch' (lose) and state 5 = 'reaches home' (win). At each step the walker moves right with probability $p = 2/3$ and left with probability $q = 1/3$.

- Write the recurrence for $p_k = P(\text{reach } 5 \mid \text{start } k)$ for the biased walk. Show that $p_k = A + B \cdot (q/p)^k$ is the general solution when $p \neq q$, noting that $q/p = 1/2$.
- Apply boundary conditions $p_0 = 0$ and $p_5 = 1$ to find A and B . Write p_k in closed form.
- Compute p_2 . Compare with the unbiased result $p_2 = 2/5$. By how much does the rightward drift increase Alice's win probability from position 2?